Experiment 2:

AIM: Queries (along with sub-Queries) using ANY, ALL, IN, EXISTS, NOTEXISTS, UNION, INTERSET, Constraints. Example: - Select the roll number and name of the student who secured fourth rank in the class.

Query(Select): A query is command to get data from a table

<u>Syntax:</u>

```
select column1, column2, ...
from table_name
where condition;
```

SubQuery(Query inside Query): A **subquery** is a query **inside another query**, the subquery is used to return data that will be used in the main query.

<u>Syntax:</u>

```
select column1, column2, ...
from table_name
where column operator (
   select column
   from table_name
   where condition
);
```

Query: Write a query to create a dept table with constraints

```
SQL>create table dept (
  deptno number primary key,
  dname varchar2(20)
);
```

OUTPUT:?

Query: write a query to insert data into dept

```
SQL>insert into dept values (10, 'IT');
SQL>insert into dept values (20, 'CSE');
SQL>insert into dept values (30, 'ECE');
```

OUTPUT:?

Query: write a query to display all data of dept table

```
SQL> select * from dept;
```

OUTPUT:?

Query: Write a query to create a student table with constraints.

```
SQL> create table student (
  rollno number primary key,
  name varchar2(20) not null,
  marks number check (marks between 0 and 100),
  deptno number,
  foreign key (deptno) references dept(deptno)
);
```

OUTPUT:?

Query: write a query to insert data into student table

```
SQL>insert into student values (1, 'ravi',95, 10);
SQL>insert into student values (2, 'sita',88, 10);
```

SQL>insert into student values (3, 'kiran',82, 20);
SQL>insert into student values (4, 'anil', 75, 20);
SQL>insert into student values (5, 'ram',65, 30);
SQL>insert into student values (6, 'john',55, 30);
SQL>insert into student values (7, 'tom',45, null);
SQL>insert into student values (8, 'lisa' 35, null);

Query: write a query to display all data of student table

SQL>select * from student;

OUTPUT:?

<u>Operators:</u>

IN

<u>Syntax:</u>

select column_name
from table_name
where column_name in (value1, value2, ...);

Query: Write a query to display roll number and name of students whose marks are 75, 82, or 95.

SQL> select rollno, name
from student
where marks in (75, 82, 95);

OUTPUT:?

ANY

Syntax:

select column_name
from table_name
where column_name operator any (subquery);

Query: Write a query to display students who scored more marks than any student who scored below 50.

SQL> select rollno, name
from student
where marks > any (
 select marks
 from student
 where marks < 50
);

OUTPUT:?

ALL

Syntax: select column_name
from table_name
where column_name operator all (subquery);

Query: Write a query to display students who scored more marks than all students who scored below 90.

SQL> select rollno, name
from student
where marks > all (
 select marks
 from student
 where marks < 90

);
OUTPUT:?

EXISTS

Syntax:

select column_name
from table_name
where exists (subquery);

Query: Write a query to display roll number and name of students who belong to a department using EXISTS

```
SQL> select rollno, name
from student s
where exists (
  select *
  from dept d
  where s.deptno = d.deptno
);
```

OUTPUT:?

NOT EXISTS

Syntax:

select column_name
from table_name
where not exists (subquery);

Query: Write a query to display students who do not belong to any department.

```
select rollno, name
from student s
where not exists (
  select *
  from dept d
  where s.deptno = d.deptno
);
```

OUTPUT:?

UNION

Syntax:

select column_name from table1
union
select column_name from table2;

Query: Write a query to display students who scored above 80 or below 40.

```
SQL>select rollno from student where marks > 80
union
select rollno from student where marks < 40;
```

OUTPUT:?

INTERSECT

Syntax:

select column_name from table1
intersect
select column_name from table2;

Query: Write a query to display students who scored above 60 and below 90.

```
select rollno from student where marks > 60
intersect
select rollno from student where marks < 90;
```

OUTPUT:?

FOURTH RANK (Subquery)

Syntax:

```
select column
from table
where column = (
 select max(column)
 from table
 where column < previous_max
);
```

Query: Write a query to display roll number and name of the student who secured fourth rank.

```
SQL> select rollno, name
from student
where marks = (
 select max(marks)
 from student
 where marks < (
  select max(marks)
  from student
  where marks < (
   select max(marks)
   from student
   where marks < (select max(marks) from student)
  )
 )
);
```

OUTPUT:?